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Abstract: This project develops a web-based Natural Language Processing (NLP) system to automate the early review of financial documents. It extracts, summarises, and analyses unstructured data while flagging anomalies using a traffic-light risk model. Results show improved efficiency, accuracy, and consistency compared to manual financial document analysis.

Introduction

Financial document review is a critical but time-consuming task, often involving large volumes of unstructured data in formats such as PDFs and spreadsheets. Manual analysis can lead to inefficiencies, inconsistencies, and increased risk of human error. This project explores the use of Natural Language Processing (NLP) to automate the early-stage review process. By extracting, analysing, and summarising key information, the system aims to support professionals in identifying insights and potential anomalies more efficiently. The goal is to enhance accuracy, reduce workload, and improve overall decision-making in financial analysis.

Financial documents are often reviewed manually, making the process slow, inefficient, and difficult to scale.

Important information is buried in the unstructured formats, making extraction and analysis challenging..

Manual analysis increases the chances of missed anomalies and inconsistencies, impacting decision-making.

Potential risks and irregularities are not always identified at the early stages, delaying investigation and response.

Methods

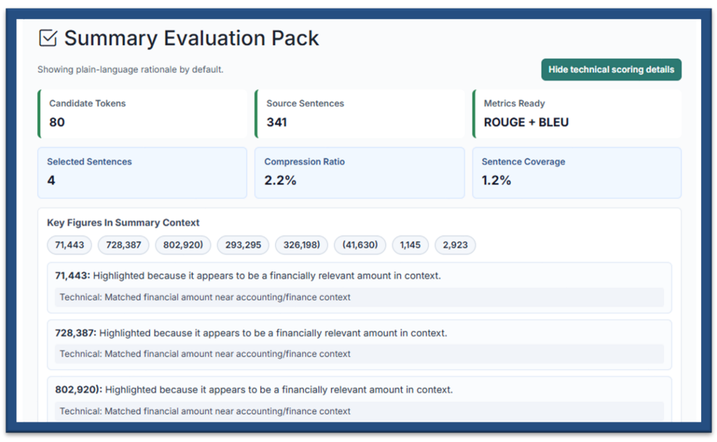
The system processes financial documents through a multi-stage NLP pipeline. Text is first extracted from PDF files using parsing and OCR techniques where required. The extracted content is then preprocessed through tokenisation, stopword removal, and lemmatisation to improve data quality.Key information is identified using Named Entity Recognition (NER), while topic modelling techniques such as TF-IDF and LDA are applied to detect underlying themes. Extractive summarisation is performed using TextRank to generate concise document summaries. Finally, a rule-based module analyses the processed data to detect anomalies and assign a traffic-light risk rating. System performance is evaluated using ROUGE and BLEU for summarisation, and precision, recall, and accuracy for anomaly detection.

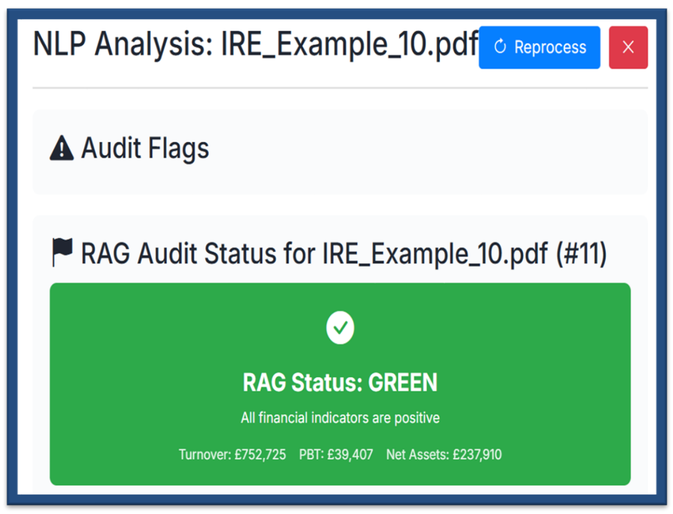
Results

The developed system successfully demonstrated the feasibility of using NLP techniques for early-stage financial document review. The text extraction module performed reliably across multiple file formats, with OCR fallback improving robustness when handling scanned or low-quality documents. Named Entity Recognition (NER) effectively identified key financial entities, while the summarisation component produced coherent summaries with acceptable ROUGE and BLEU scores, indicating good alignment with reference outputs.

The anomaly detection module was able to flag irregularities using a rule-based approach, providing clear traffic-light classifications (Green, Amber, Red) to assist users in prioritising further investigation. Integration between the frontend and backend systems functioned as expected, with API responses and processing pipelines operating within acceptable performance thresholds.

Overall, the system met its core objectives by reducing manual effort, improving consistency in document analysis, and demonstrating strong potential for scalability and future enhancement.





Conclusion

This project demonstrates how Natural Language Processing (NLP) can be applied to automate the early-stage review of financial documents. By extracting, summarising, and analysing unstructured data, the system reduces reliance on manual processes and improves efficiency and consistency. The integration of anomaly detection provides additional support for identifying potential risks. Overall, the results highlight the potential of NLP-driven solutions to enhance financial analysis workflows, with scope for future improvements through more advanced models and real-world deployment.

Literature Cited

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